Evaluating Deep Learning Pipelines for Predicting Electricity Prices in Renewable-Dominated Markets

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Abstract

Accurate electricity price forecasting is essential for market participants navigating increasingly renewable-dominated grids, where intermittent generation introduces high price volatility and non-linear temporal dynamics. This study compares end-to-end and multi-stage forecasting pipelines for day-ahead and real-time price prediction for the electricity market in Spain, evaluating both hybrid CNN+BiLSTM with attention and a standard Transformer architectures. Using publicly available data for 2015-2018, we find that the end-to-end pipeline outperforms the multi-stage pipeline with approximately 50% lower MAE and MAPE across both forecasting horizons, with the Transformer achieving the best day-ahead performance and CNN+BiLSTM achieving the best real-time performance.

1 Introduction

The modern energy sector has undergone an evolution from vertically-integrated monopolies which controlled all aspects of power delivery, including generation, transmission, and distribution to the current deregulated system of energy markets, which has created competitive positions for generators, distributors, and consumers Castelli et al. [2020]. Pricing mechanisms play a fundamental role in setting competitive market bids, ensuring revenue generation, which in turn supports power grid stability. Therefore, electricity price forecasting serves as an invaluable tool for market participants to optimize and negotiate their trading and bidding strategies. Accurate predictions of short- and long-term electricity prices help manage supply and demand in real-time and reduce financial risks for market participants by allowing them to hedge against price volatility Montel [2025]. However, increasing shares of renewable generation sources make forecasting prices particularly challenging due to their intermittency and lower marginal costs. The competitive bidding behaviors of market participants, operating conditions of the power system and the increased penetration of renewables all present a dynamic landscape where forecasting electricity prices becomes highly non-linear.

Electricity price forecasting comprises short-term and long-term forecasting. While long-term forecasting is typically used for infrastructure planning and investment decisions, short-term forecasting is typically used to predict real-time or day-ahead energy prices which are particularly relevant for maintaining grid stability and facilitating energy trading Montel [2025]. The real-time market - which consists of the Intra-Day and Balancing Markets - have high temporal resolutions with market transactions occurring within 15 minutes to several hours ahead of supply delivery. The high-frequency nature of these markets makes them particularly sensitive to fluctuations in renewable energy generation as inaccurate forecasting hinders managing supply-demand imbalances O'Connor et al. [2025]. In contrast, the day-ahead market operates using a forward scheduling mechanism with a 24 hour temporal resolution in which generators commit their electricity trades the day before delivery. Prices in the day-ahead market are determined through forecasted supply and demand. However, the variability of renewables and their dependence on weather conditions can make forecasting a challenge.

Statistical methods, including Autoregressive Integrated Moving Average (ARIMA) or Generalized AutoRegressive Conditional Heteroskedasticity (GARCH), have traditionally been used for short-term price forecasting to capture temporal dependencies in electricity markets and seasonality in generation supply and load. However, regression-based models rely on linearity and stationarity and may struggle to capture non-linear dependencies, dynamic patterns, and external factors associated with electricity prices including weather conditions, consumption patterns, and fuel prices. Machine learning methods such as Support Vector Machines (SVM), Random Forests (RF), Gradient Boosting Machines (GBM), and XGBoost have emerged as an alternative for improving accuracy and capturing non-linear relationships between weather, renewable generation, grid demand, and peak loads, particularly under rapidly changing operating conditions O'Connor et al. [2025]. Recently, deep learning methods have been explored to improve forecasting accuracy due to their ability to capture non-linear dependencies between inputs and outputs, hierarchically learn features that influence market prices, and efficiently extract patterns from large-scale data Zhang et al. [2020]. Recurrent Neural Networks (RNN) are particularly effective at representing and interpreting dynamic and sequential data which makes them suitable for time-series based electricity data Zhang et al. [2020]. However, conventional RNN's encounter problems of capturing long-term dependencies due to vanishing gradients. Long Short-Term Memory (LSTM) models address this gap by preserving temporal dependencies through their internal structure where they can selectively retain and forget historical information. Convolutional Neural Networks (CNNs) are also relevant for being able to capture local features and periodic fluctuations in power systems and market operations. Hybrid architectures such as CNN-LSTM are suitable for improving time-series forecasting by combining convolutional feature extraction in the short-term with recurrent sequence learning in the long-term. In addition, transformer-based architectures have recently gained attention due to their ability to model long-range dependencies through self-attention mechanisms. Several studies have reported that Transformer models outperform RNNs for electricity price forecasting, particularly when handling large datasets and complex temporal patterns Zhong [2023].

Deep learning methods have been extensively studied for day-ahead electricity price forecasting due to its longer forecasting horizon and lower dependence on high-frequency exogenous variables O'Connor et al. [2025]. Similar approaches have also been applied to renewable generation and electricity demand forecasting, with studies exploring joint demand-price forecasting to capture their dynamic interdependence Nafea et al. [2025]. In contrast, the high-frequency and volatile nature of intra-day markets makes real-time price forecasting considerably challenging with a sparsity in studies investigating deep learning approaches. Furthermore, while a large body of work exists to compare forecasting model architectures, limited work exists in examining how different forecasting pipelines influence predictive performance.

To address these gaps, this study evaluates deep learning forecasting pipelines for electricity price prediction in a renewable-dominated electricity market. We compare a direct, end-to-end forecasting pipeline — which takes historical generation, load, and system operator forecasts of renewable generation and demand as inputs — against a multi-stage pipeline that sequentially generates those intermediate forecasts before predicting market prices, mirroring realistic market operations Hua et al. [2025]. Both pipelines predict real-time and day-ahead electricity prices using CNN-BiLSTM and Transformer architectures. Our primary contribution is assessing the suitability of end-to-end versus multi-stage forecasting strategies for ensuring greater predictive accuracy. This work provides insights into how renewable generation and demand dynamics should be incorporated into deep learning-based price forecasting methods for robust applications across different market conditions.

2 Methodology

We developed two distinct forecasting pipelines to predict day-ahead and real-time prices in an energy market with a high share of renewables in the generation mix. An end-to-end forecasting pipeline directly predicts the day-ahead and real-time prices using generation, system load, and published forecasts. Meanwhile, a multi-stage pipeline follows a sequential prediction schema where weather data is initially used to forecast renewable generation and system load which are subsequently used along with additional generation, temporal, and published real-time forecasts to predict market prices. Historical time-series for generation, energy consumption, market settlement prices, and

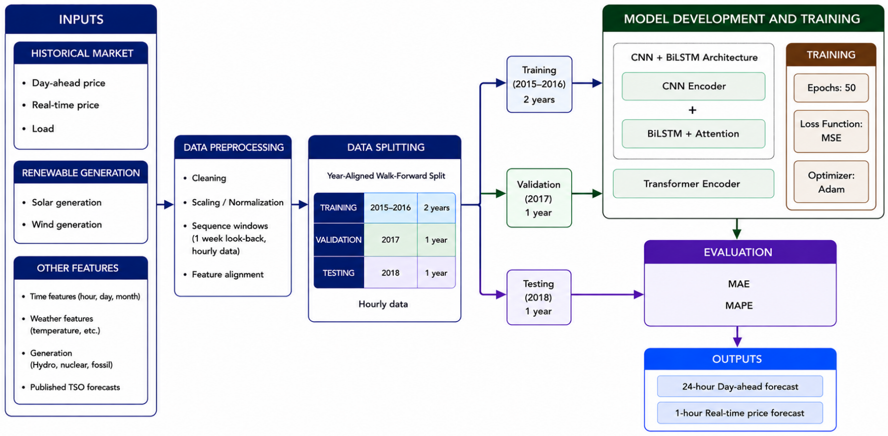

Figure 1: Data processing, model development, and evaluation workflow.

weather features served as inputs into two model architectures: a hybrid convolutional network with bidirectional LSTM and a standard Transformer encoder. We compared results across both architectures for the end-to-end and multi-stage forecasting pipelines. We conducted all experiments for both pipelines using Python's PyTorch library on a single NVIDIA L40S GPU. All models were trained for 50 epochs with an initial learning rate set at 10^{-3}. We used the Adaptive Moment Estimation (Adam) optimizer to ensure efficient memory usage and dynamically adapt the learning rate for each parameter. Our experiment methodology is illustrated in Figure 1, and all code is publicly available at `https://github.com/amukherjee25/forecasting-electricity-prices`.

2.1 Dataset

We used Spain as a case study to represent an energy market with a high penetration of renewables. Spain's generation mix and installed capacity accounts for over 38% renewables with wind generation consisting over 19% of the overall generation mix (Figure 2). We obtained data on generation by technology, system load, market prices, and weather spanning across 2015 - 2018 at an hourly resolution. The European Network of Transmission System Operators for Electricity (ENTSO-E) publishes generation and consumption data. We obtained settlement prices, which included both day-ahead and real-time market settlement prices, from Spain's Transmission System Operator (TSO), Red Electric España. Weather data was published on Open Weather API and included temperature, precipitation, humidity, pressure, snowfall, and cloud cover for five cities in Spain: Bilbao, Barcelona, Madrid, Seville, and Valencia. Final target variables for both forecasting pipelines included day-ahead prices for each hour of the subsequent day representing Day Ahead Market (DAM) prices and the hour ahead price for the current day, representing the IntraDay Market (IDM). We used a walk-forward cross validation fold of two years of data (2015-2016) for model training, one year of data (2017) for validation, and the final year of data (2018) for model testing and evaluation. The end-to-end forecasting pipeline used a 48-hour input sequence (i.e., two days of historical data) to predict both day-ahead and real-time prices. In contrast, the multi-stage pipeline used a 168-hour input sequence (one week) to capture weekly patterns in energy consumption and generation.

2.2 Feature Engineering & Selection

We pre-processed all input features to ensure data quality and temporal alignment, used linear interpolation to handle missing values, and dropped features with substantial missing data. We performed a cyclical encoding of all time variables (hour, day, and month) as sine/cosine pairs to represent temporal proximity and introduced a binary weekend indicator to capture variability in load patterns. We further augmented the dataset with 1-, 2-, 3-, 24-, and 48-hour lag features for renewable generation (e.g. solar and wind), total system load, and market prices to better capture temporal dynamics. We normalized all input features and target variables using Min-Max scaling, excluding cyclical encodings which were already bounded within [-1,1].

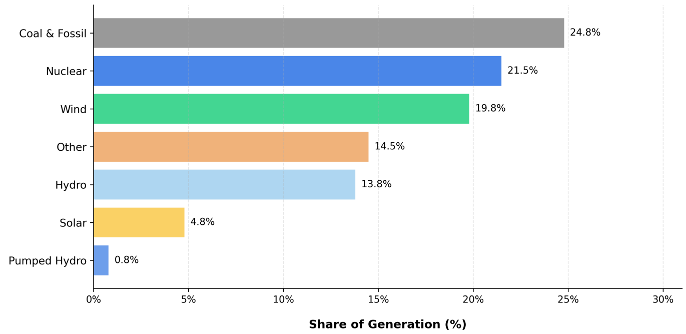

Figure 2: Spain's energy generation mix in 2018

Feature selection varied based on the relevant feature inputs for the forecasting pipeline. Weather features were not informative for the end-to-end forecasting pipeline as iterative selection processes revealed poor model predictive metrics due to higher validation losses. Therefore, these were excluded for the end-to-end forecasting pipeline. However, weather features across all five cities in Spain were included in the multi-stage forecasting pipeline as they would be vital for predicting next-day forecasts for solar and wind generation and system load. Similarly, published TSO generation and load forecasts were used as lagged input features in the end-to-end pipeline for predicting day-ahead and real-time prices. However, these features were only added in up to the previous day when predicting real-time prices in the multi-stage pipeline in order to avoid data leakage.

2.3 Model Architectures

2.3.1 CNN+BiLSTM

To capture local feature extraction alongside long-range temporal dependencies, we developed a hybrid architecture combining a one-dimensional Convolutional Neural Network (CNN) with a Bi-Directional Long Short-Term Memory (BiLSTM) and attention mechanism (Figure 3. The CNN consists of three one-dimensional convolutional layers with batch normalization and kernel sizes of 3, 8, and 12 hourly timesteps. These kernel sizes were selected to capture fine-resolution dynamics (3 timesteps) such as sudden spikes in load or price, block-level generation patterns (8 timesteps) associated with renewable generation cycles, and half-day cycles (12 timesteps) capturing both morning and evening variations and demand peaks. The CNN output is passed to a two-layer BiLSTM, which models temporal dependencies of the input features in both forward and backward directions. An attention mechanism computes a weighted sum over the BiLSTM hidden states to produce a fixed-length context vector to emphasize the most relevant timesteps of the input sequence for generating predictions. Two decoders use this context vector to generate price predictions. An autoregressive decoder predicts day-ahead prices by generating hourly scalar outputs sequentially, where the price for the predicted hour is used as input for predicting the price at the next time step. A dense decoder predicts real-time prices using a multi-layer, fully connected feed forward network with ReLU activations.

2.3.2 Transformer

To improve model efficiency and enable parallel processing of temporal features, we developed a standard Transformer encoder architecture (Figure 4 with self-attention as an alternative to the recurrent processing of the CNN+BiLSTM. The encoder takes in a fixed-length historical window of the input time-series and projects each feature vector into a 64-dimensional embedding space where sinusoidal positional encoding is applied. The encoder consists of two stacked layers, each containing a multi-head self-attention layer with 4 heads, allowing the model to jointly attend to temporal dependencies across features simultaneously, followed by a feedforward network with ReLU activation. Residual connections and layer normalization are applied after each sublayer to stabilize training and preserve gradient flow. The encoder output is aggregated via mean pooling to produce a fixed-length context vector which ensures that all historical temporal features contribute equally to the final price prediction. A two-layer dense decoder is used to predict all hourly day-ahead

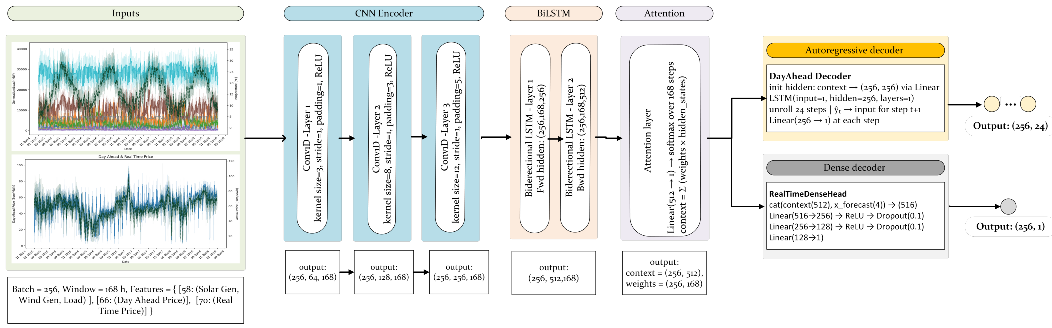

Figure 3: CNN + BiLSTM model architecture with input and output dimensions for each encoder and decoder layers.

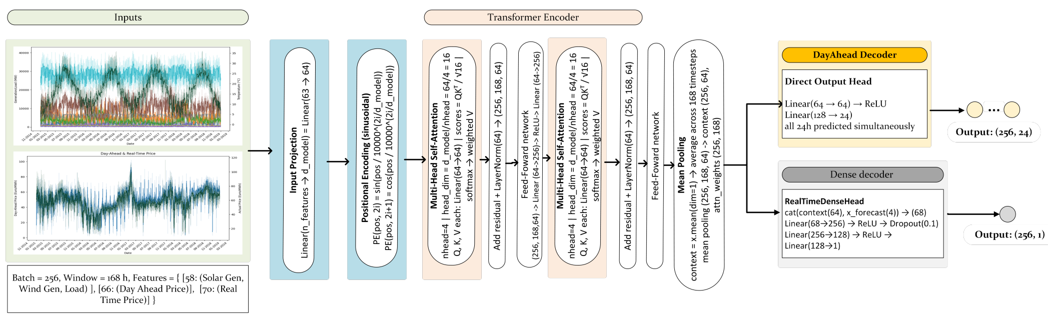

Figure 4: Transformer model architecture with input and output dimensions for each encoder and decoder layers.

prices through a direct projection since the self-attention mechanism is able to capture intra-day dynamics between generation, load, and price globally. Meanwhile, a three-layer dense decoder is used for real-time price prediction with the additional depth used to model the potentially higher volatility of real-time market prices.

2.3.3 Hyperparameter Tuning

We tuned model hyperparameters sequentially for each architecture in each pipeline. Given the longer training times required for the multi-stage forecasting pipeline, hyperparameter tuning was limited to adjusting the dropout between BiLSTM layers and the channel and kernel sizes in the convolutional layers. To further tune the model and improve its predictive performance, we would consider implementing a grid search to obtain the optimal hyperparameters across the model. Meanwhile, the faster training times of the end-to-end forecasting pipeline allowed for a more robust hyperparameter tuning. We tuned the following hyperparameters in a sequential order: dropout within the network, the input sequence length (i.e. hourly lookback on historical data), optimizer learning rate, and embedding dimension of the transformer architecture. We also performed an ablation study on the end-to-end forecasting pipeline by removing any lagged and weather features to determine their impact on the model. The ablation analysis revealed that adding in weather features into the end-to-end pipeline caused predictive performance to deteriorate. Consequently, weather features were entirely excluded from this pipeline. We used the Mean Squared Error (MSE) loss function across both pipelines and architectures as it yielded optimal results in comparison to Mean Absolute Error (MAE).

2.3.4 Evaluation Metrics

We evaluated model performance based on three metrics: Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Root Mean Squared Error (RMSE). MAE served as an indicator for the deviation between the predicted and actual day-ahead and real-time prices. Meanwhile, MAPE provided the percentage deviation between the predicted and actual values, with lower percentages indicating better model performance. Although RMSE was initially used as an evaluation metric, it was discarded due to its limited interpretability in how model predictions were performing against financial metrics. Additionally, we disaggregated prediction errors across seasons to capture forecast difficulty across shorter temporal horizons.

3 Results

The end-to-end pipeline outperformed the multi-stage pipeline across both forecasting horizons, with MAE and MAPE approximately 50% lower for both day-ahead and real-time predictions (Table 1 & Table 2).

Table 1: Day-Ahead Electricity Market Price Predictions

Pipeline	Architecture	MAE	MAPE	Training Time	Parameters
Multi-Stage	Transformer	11.54 €/MWh	31.44%	5.45 min	154,945
Multi-Stage	CNN+BiLSTM	11.00 €/MWh	31.87%	15.16 min	3,268,290
End-to-End	CNN+BiLSTM	8.72 €/MWh	20.38%	2.58 min	2,541,337
End-to-End	**Transformer**	**6.49 €/MWh**	**15.53%**	**1.65 min**	**103,064**

Table 2: Real-Time Electricity Market Price Predictions

Pipeline	Architecture	MAE	MAPE	Training Time	Parameters
Multi-Stage	Transformer	11.64 €/MWh	18.89%	5.37 min	109,976
Multi-Stage	CNN+BiLSTM	8.13 €/MWh	13.77%	16.43 min	4,158,146
End-to-End	Transformer	3.45 €/MWh	6.21%	1.62 min	101,569
End-to-End	**CNN+BiLSTM**	**2.91 €/MWh**	**5.09%**	**2.05 min**	**2,529,538**

We expected the multi-stage pipeline to have a weaker performance due its sequential structure where errors in upstream renewable generation and load forecasts can propagate into downstream price predictions. Within the end-to-end pipeline, the Transformer excelled in predicting day-ahead prices while the CNN+BiLSTM performed better on real-time prices, consistent with the structural differences between both architectures. For day-ahead forecasting, the Transformer's self-attention mechanism enables integration of broader temporal context across the full 24-hour horizon, allowing it to better capture intra-day price trends (Figure 5).

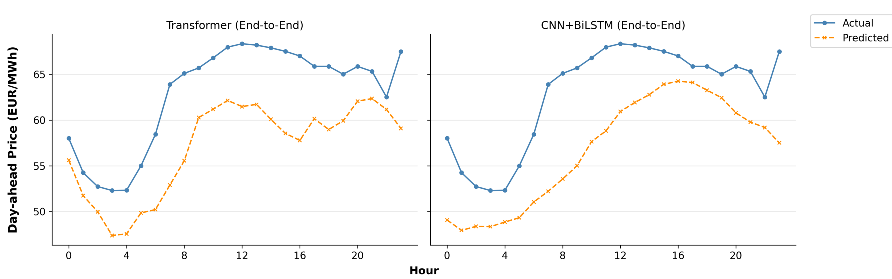

Figure 5: Predicted vs. actual day-ahead prices for a representative day using the end-to-end pipeline: Transformer (left) and CNN+BiLSTM (right).

For real-time forecasting, the CNN+BiLSTM's local feature extraction coupled with bidirectional sequential processing is better suited to the short-horizon, high-frequency dynamics of settlement prices (Figure 6). Across both tasks, the end-to-end architectures exhibited smaller prediction errors and closer tracking of actual prices compared to their multi-stage counterparts.

Additionally, we found that seasonal variability in renewable generation also impacts model predictive potential. Wind generation accounts for over 19% of Spain's energy mix and demonstrates the highest variability during the spring season due to the Scandinavian wind patterns (SCAND) which are associated with above-normal wind events Bayo-Besteiro et al. [2022]. This variability is reflected in

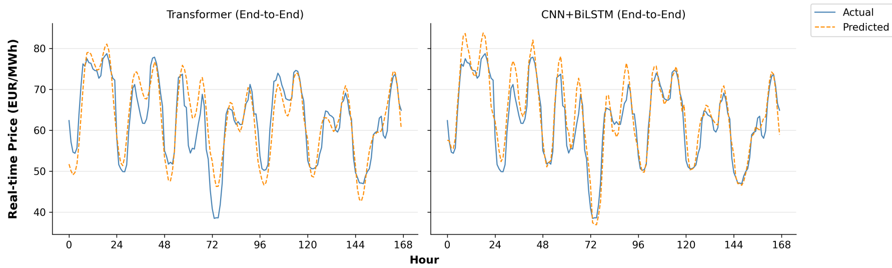

Figure 6: Predicted vs. actual real-time prices for a representative week using the end-to-end pipeline: Transformer (left) and CNN+BiLSTM (right).

both day-ahead and real-time market prices, which also demonstrate the highest variations during spring. These characteristics in turn correspond to the lowest predictive performance during spring in comparison to other seasons (Figure 7). This seasonal degradation in model performance suggests the need for alternative forecasting models and strategies specifically for utilities, transmission system operators, and other market participants for whom forecasting reliability is crucial for operational decisions and revenue generation.

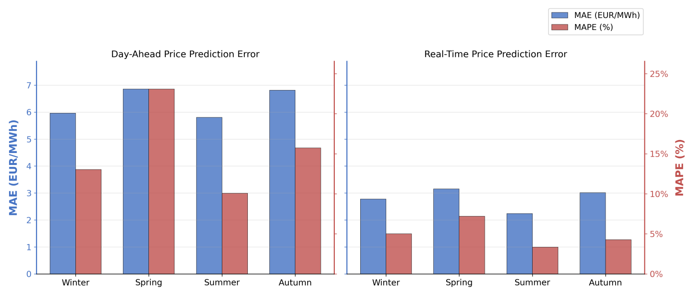

Figure 7: Seasonal prediction error for day-ahead and real-time electricity prices. MAE (EUR/MWh) and MAPE (%) across all seasons for the best-performing pipeline & architectures: Transformer (day-ahead) and CNN+BiLSTM (real-time)

4 Conclusion

This study evaluated end-to-end and multi-stage deep learning forecasting pipelines for day-ahead and real-time electricity price prediction in a renewable-dominated market. We found that the end-to-end pipeline outperformed the multi-stage pipeline across both horizons, with the Transformer architecture being best suited for day-ahead forecasting and CNN+BiLSTM for real-time forecasting, reflecting the alignment between each architecture's inductive biases and the temporal characteristics of each market. These results suggest that directly incorporating predicted renewable generation and demand dynamics as inputs rather than sequentially forecasting them yields more accurate price predictions. Nonetheless, the multi-stage pipeline would likely benefit from additional meteorological and energy consumption data over longer temporal spans and broader geographical coverage to improve model robustness and reduce error propagation across sequential forecasting stages. Additionally, while the CNN+BiLSTM architecture demonstrated strong performance on real-time price forecasting, its bidirectional nature relies on future context unavailable in operational settings. Future work will explore an unidirectional LSTM to better reflect the causal constraints of real-time market forecasting. Beyond model architecture, our analysis also highlighted the critical role of renewable generation in shaping electricity prices. Wind generation emerged as a particularly influential predictor across both pipelines, and its high seasonal variability, especially during spring, was directly reflected in higher model prediction errors during that period. As renewable penetration continues to increase European electricity markets, improvements in forecasting would lead to more accurate price signals in both day-ahead and real-time horizons, resulting in financial benefits for all market participants.

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